

Program No.17

Aim:

To build a basic Local Area Network (LAN) in Cisco Packet Tracer, you connect multiple computers to a central switch and assign static IPv4 addresses within the same subnet

Requirements and Equipment

- Software: Cisco Packet Tracer
- End Devices: 3 to 4 PCs or Laptops
- Network Device: 1 Switch (e.g., model 2960)
- Cables: Copper Straight-Through cables

Step-by-Step Experiment Procedure

1. Place Devices on Workspace

- o Open Cisco Packet Tracer.
- o Click **End Devices** at the bottom-left menu and drag three PCs (PC0, PC1, PC2) onto the logical workspace.
- o Click **Switches** and drag a Switch (Switch0) to the center of the workspace.

2. Connect the Devices

- Select the **Connections** icon (lightning bolt symbol) and pick the black diagonal line for **Copper Straight-Through** cable.
- Connect each PC to Switch0:
 - o Click PC0 → choose **FastEthernet0** → connect to Switch0 port **FastEthernet0/1**.
 - o Click PC1 → choose **FastEthernet0** → connect to Switch0 port **FastEthernet0/2**.
 - o Click PC2 → choose **FastEthernet0** → connect to Switch0 port **FastEthernet0/3**.
- Wait a few seconds until the link indicator lights on the cable ends turn **green**, indicating an active connection.

3. Configure IP Addresses

- Click on PC0, go to the **Desktop** tab, and open **IP Configuration**.
- Select **Static** and enter the IPv4 configuration details:
 - o **IP Address:** 192.168.1.10
 - o **Subnet Mask:** 255.255.255.0
- Repeat this process for the remaining PCs, incrementing the final octet of the IP address:
 - o PC1: IP Address 192.168.1.11, Subnet Mask 255.255.255.0
 - o PC2: IP Address 192.168.1.12, Subnet Mask 255.255.255.0

4. Verify Connectivity (Ping Test)

- Click on PC0, open the **Desktop** tab, and click **Command Prompt**.
- Test the connection to PC1 by typing the following command and pressing Enter:

```
ping 192.168.1.11
```

Confirm that replies are received with 0% packet loss. You can similarly ping 192.168.1.12 from PC0 to verify full local connectivity.